ex. 1) Aruncarea unei monede → H = succes

$X: \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ (0 ↔ exec, 1 ↔ succes)

$X \sim \text{Unif}(\{0,1\})$

$X \sim \text{Bern}(\frac{1}{2})$

2) Aruncarea unui zar

$X: \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \end{pmatrix}$

$X \sim \text{Unif}(\{1,2,3,4,5,6\})$

aproximare cu normala standard

$F(x) = P(X \le x) = \begin{cases} 0, & x < 1 \\ \frac{1}{6}, & 1 \le x < 2 \\ \frac{2}{6}, & 2 \le x < 3 \\ \frac{3}{6}, & 3 \le x < 4 \\ \frac{4}{6}, & 4 \le x < 5 \\ \frac{5}{6}, & 5 \le x < 6 \\ 1, & x \ge 6 \end{cases}$

nr. de succese din n încercări independente cu prob. de succes p

funcția gama incompletă

$X: \begin{pmatrix} 0 & 1 \\ 1-p & p \end{pmatrix}$

C_n^x

p.m.f.

$\mathbb{1}_{(a,b)}(x)$

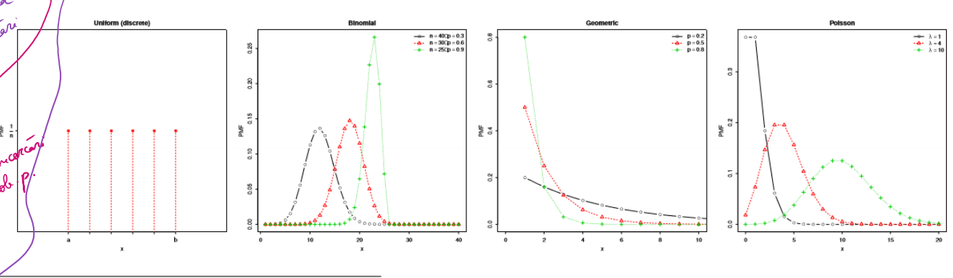
funcție generatoare de momente

1 Distribution Overview

1.1 Discrete Distributions

prob. de succes

	Notation[1]	$F_X(x)$	$f_X(x)$	$\mathbb{E}[X]$	$\mathbb{V}[X]$	$M_X(s)$
Uniform	$\text{Unif}\{a,\dots,b\}$	$\begin{cases} 0 & x < a \\ \frac{\lfloor x \rfloor - a + 1}{b-a+1} & a \le x \le b \\ 1 & x > b \end{cases}$	$\frac{I(a < x < b)}{b-a+1}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$	$\frac{e^{as} - e^{-(b+1)s}}{s(b-a)}$
Bernoulli	$\text{Bern}(p)$	$(1-p)^{1-x}$ $x \in \{0,1\}$	$p^x(1-p)^{1-x}$	p	$p(1-p)$	$1-p+pe^s$
Binomial	$\text{Bin}(n,p)$	$I_{1-p}(n-x,x+1)$	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$	$(1-p+pe^s)^n$
Multinomial	$\text{Mult}(n,p)$		$\frac{n!}{x_1! \dots x_k!} p_1^{x_1} \cdots p_k^{x_k}$ $\sum_{i=1}^k x_i = n$	np_i	$np_i(1-p_i)$	$\left(\sum_{i=0}^k p_i e^{s_i}\right)^n$
Hypergeometric	$\text{Hyp}(N,m,n)$	$\approx \Phi\left(\frac{x-np}{\sqrt{np(1-p)}}\right)$	$\frac{\binom{m}{x}\binom{m-x}{n-x}}{\binom{N}{n}}$	$\frac{nm}{N}$	$\frac{nm(N-n)(N-m)}{N^2(N-1)}$	
Negative Binomial	$\text{NBin}(r,p)$	$I_p(r,x+1)$	$\binom{x+r-1}{r-1} p^r (1-p)^x$	$r\frac{1-p}{p}$	$r\frac{1-p}{p^2}$	$\left(\frac{p}{1-(1-p)e^s}\right)^r$
Geometric	$\text{Geo}(p)$	$1-(1-p)^x$ $x \in \mathbb{N}^+$	$p(1-p)^{x-1}$ $x \in \mathbb{N}^+$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{p}{1-(1-p)e^s}$
Poisson	$\text{Po}(\lambda)$	$e^{-\lambda} \sum_{i=0}^x \frac{\lambda^i}{i!}$	$\frac{\lambda^x e^{-\lambda}}{x!}$	λ	λ	$e^{\lambda(e^s-1)}$

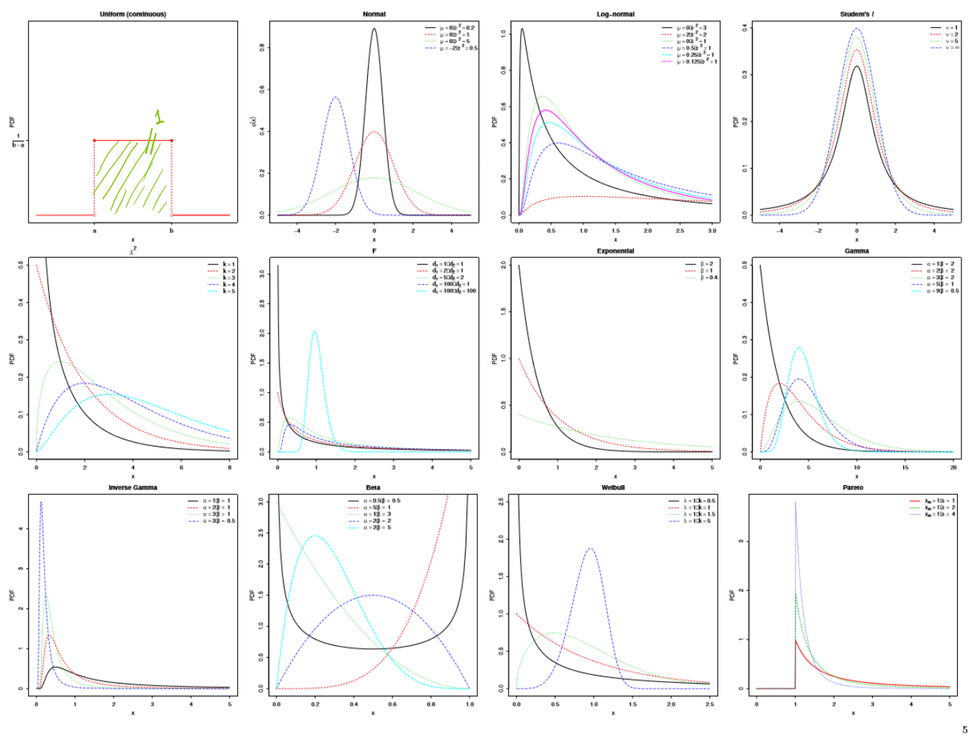


[1] We use the notation $\gamma(s,x)$ and $\Gamma(x)$ to refer to the Gamma functions (see §22.1), and use $B(x,y)$ and I_x to refer to the Beta functions (see §22.2).

nr. de apariții ale fenomenului ☺ știind că nr. mediu de apariții este λ într-un interval dat de timp în care fenomenul este uniform distribuit

1.2 Continuous Distributions

	Notation	$F_X(x)$	$f_X(x)$	$\mathbb{E}[X]$	$\mathbb{V}[X]$	$M_X(s)$
Uniform	$\text{Unif}(a,b)$	$\begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a < x < b \\ 1 & x > b \end{cases}$	$\frac{I(a < x < b)}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{sb} - e^{sa}}{s(b-a)}$
Normal	$\mathcal{N}(\mu,\sigma^2)$	$\Phi(x) = \int_{-\infty}^x \phi(t)\,dt$	$\phi(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	μ	σ^2	$\exp\left\{\mu s + \frac{\sigma^2 s^2}{2}\right\}$
Log-Normal	$\ln\mathcal{N}(\mu,\sigma^2)$	$\frac{1}{2} + \frac{1}{2}\text{erf}\left[\frac{\ln x - \mu}{\sqrt{2\sigma^2}}\right]$	$\frac{1}{x\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(\ln x - \mu)^2}{2\sigma^2}\right\}$	$e^{\mu+\sigma^2/2}$	$(e^{\sigma^2}-1)e^{2\mu+\sigma^2}$	
Multivariate Normal	$\text{MVN}(\mu,\Sigma)$		$(2\pi)^{-k/2}\lvert\Sigma\rvert^{-1/2} e^{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)}$	μ	Σ	$\exp\left\{\mu^T s + \frac{1}{2}s^T\Sigma s\right\}$
Student's t	$\text{Student}(\nu)$	$I_x\left(\frac{\nu}{2},\frac{\nu}{2}\right)$	$\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-(\nu+1)/2}$	0	0	
Chi-square	χ^2_k	$\frac{1}{\Gamma(k/2)}\gamma\left(\frac{k}{2},\frac{x}{2}\right)$	$\frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$	k	$2k$	$(1-2s)^{-k/2}$ $s < 1/2$
F	$\text{F}(d_1,d_2)$	$I_{\frac{d_1 x}{d_1 x + d_2}}\left(\frac{d_1}{2},\frac{d_1}{2}\right)$	$\frac{\sqrt{\frac{(d_1 x)^{d_1} d_2^{d_2}}{(d_1 x + d_2)^{d_1+d_2}}}}{x\text{B}\left(\frac{d_1}{2},\frac{d_1}{2}\right)}$	$\frac{d_2}{d_2-2}$	$\frac{2d_2^2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)}$	
Exponential	$\text{Exp}(\beta)$	$1-e^{-x/\beta}$	$\frac{1}{\beta}e^{-x/\beta}$	β	β^2	$\frac{1}{1-\beta s}$ $(s < 1/\beta)$
Gamma	$\text{Gamma}(\alpha,\beta)$	$\frac{\gamma(\alpha,x/\beta)}{\Gamma(\alpha)}$	$\frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}$	$\alpha\beta$	$\alpha\beta^2$	$\left(\frac{1}{1-\beta s}\right)^\alpha$ $(s < 1/\beta)$
Inverse Gamma	$\text{InvGamma}(\alpha,\beta)$	$\frac{\Gamma\left(\alpha,\frac{\beta}{x}\right)}{\Gamma(\alpha)}$	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} e^{-\beta/x}$	$\frac{\beta}{\alpha-1}$ $\alpha > 1$	$\frac{\beta^2}{(\alpha-1)^2(\alpha-2)^2}$ $\alpha > 2$	$\frac{2(-\beta s)^{\alpha/2}}{\Gamma(\alpha)} K_\alpha\left(\sqrt{-4\beta s}\right)$
Dirichlet	$\text{Dir}(\alpha)$		$\frac{\Gamma\left(\sum_{i=1}^k \alpha_i\right)}{\prod_{i=1}^k \Gamma(\alpha_i)} \prod_{i=1}^k x_i^{\alpha_i - 1}$	$\frac{\alpha_i}{\sum_{i=1}^k \alpha_i}$	$\frac{\mathbb{E}[X_i](1-\mathbb{E}[X_i])}{\sum_{i=1}^k \alpha_i + 1}$	
Beta	$\text{Beta}(\alpha,\beta)$	$I_x(\alpha,\beta)$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$1 + \sum_{k=1}^\infty \left(\prod_{r=0}^{k-1} \frac{\alpha+r}{\alpha+\beta+r}\right) \frac{s^k}{k!}$
Weibull	$\text{Weibull}(\lambda,k)$	$1-e^{-(x/\lambda)^k}$	$\frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}$	$\lambda\Gamma\left(1+\frac{1}{k}\right)$	$\lambda^2\Gamma\left(1+\frac{2}{k}\right) - \mu^2$	$\sum_{n=0}^\infty \frac{s^n\lambda^n}{n!}\Gamma\left(1+\frac{n}{k}\right)$
Pareto	$\text{Pareto}(x_m,\alpha)$	$1-\left(\frac{x_m}{x}\right)^\alpha$ $x \ge x_m$	$\alpha\frac{x_m^\alpha}{x^{\alpha+1}}$ $x \ge x_m$	$\frac{\alpha x_m}{\alpha-1}$ $\alpha > 1$	$\frac{x_m^\alpha}{(\alpha-1)^2(\alpha-2)}$ $\alpha > 2$	$\alpha(-x_m s)^\alpha \Gamma(-\alpha, -x_m s)$ $s < 0$



ex. Dintr-o urnă cu 15 bile roșii și 10 bile negre se extrag, cu revenire, 3 bile. Fie X nr. bile roșii

a) Construiți v.a. X și F

b) Calc. $P(X > 1 \mid X > 0)$

$X \sim \text{Bin}(3, \frac{15}{25} = \frac{3}{5})$

$X: \begin{pmatrix} 0 & 1 & 2 & 3 \\ p_0 & p_1 & p_2 & p_3 \end{pmatrix}$

$p_k = C_3^k \cdot \left(\frac{3}{5}\right)^k \cdot \left(1 - \frac{3}{5}\right)^{3-k}$ $\forall k \in \{0,1,2,3\}$

$F(x) = \begin{cases} 0, & x < 0 \\ p_0, & 0 \le x < 1 \\ p_0 + p_1, & 1 \le x < 2 \\ p_0 + p_1 + p_2, & 2 \le x < 3 \\ 1, & x \ge 3 \end{cases}$

b) $P(X > 1 \mid X > 0) = \frac{P((X>1) \cap (X>0))}{P(X>0)} = \frac{P(X>1)}{P(X>0)} = \frac{1 - P(X \le 1)}{1 - P(X \le 0)}$

$= \frac{1 - (p_0 + p_1)}{p_0}$

timpul de așteptare până la prima apariție a fenomenului ☺ știind că nr. mediu de apariții este λ

$\lambda = \frac{1}{\beta}$

același λ ca la Poisson

$f(x) = \lambda \cdot e^{-\lambda x} \cdot \mathbb{1}_{(0,\infty)}(x)$